MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

Discipline-Specific Elective Course (DSE)

MG4DSEBOTA06: Environmental and Industrial Microbiology in Agriculture

(2024 ADMISSION ONWARDS)

Time: 2 Hrs. **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I), and Appreciation (Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

One or two Sentences

Answer any ten questions. Each question carries 2 marks

1. What is an agroecosystem?	[K]	[1]
2. Mention any two advantages of microbial mitigation of agrochemicals.	[U]	[2]
3. Mention two microbes involved in the nitrogen cycle.	[K]	[2]
4. What is biogas?	[K]	[2]
5. Name any two products obtained through agro-industrial fermentation.	[K]	[4]
6. Write any two advantages of in situ bioremediation.	[U]	[2]
7. Differentiate between submerged and solid-state fermentation.	[U]	[3]
8. Identify the fermentation process involved in wine production and justify.	[A]	[3]
9. How does *Rhizobium* benefit leguminous plants?	[U]	[3]
10. Identify the major microbial role of Beejamrutha in seed health.	[U]	[6]
11. Why are indigenous biocontrol practices considered sustainable?	[U]	[6]
12. What is Pokkali rice cultivation?	[K]	[5]

(2 × 10 = 20)

PART - B

Short answer

Answer any eight questions. Each question carries 5 marks

13. Explain the role of microorganisms in the biodegradation of pesticides and heavy metals in agricultural soils.	[U]	[2]
14. Describe the role of microorganisms in nitrogen and phosphorus cycles in agroecosystems.	[U]	[2]
15. Describe the production of biochar and organic acids from agricultural wastes.	[U]	[4]
16. Explain microbial indicators used to assess soil and water quality in agriculture.	[U]	[1]
17. Illustrate how *Azotobacter* and phosphate-solubilizing bacteria can be applied as biofertilizers to improve crop productivity.	[A]	[3]
18. What is meant by indigenous biocontrol practices? List any five traditional microbial agents used in pest or disease control.	[K]	[6]
19. What are plant growth-promoting substances? List the microbial sources of gibberellins and indole-3-acetic acid (IAA).	[K]	[3]
20. Evaluate the environmental and economic benefits of using agro-wastes for bioenergy and organic acid production.	[E]	[4]
21. Analyse the Kaipad rice system as a model of rice–fish integrated farming.	[An]	[5]
22. Define microbial antagonism and list its types observed in traditional pest management.	[K]	[6]

(5 × 8 = 40)

PART – C

Essay

Answer any one question. Each question carries 10 marks

23. Describe alcoholic and lactic acid fermentations and give examples of their applications in the food and beverage industries.	[U]	[3]
24. Explain the traditional farming systems of Kerala. Analyze their ecological and socio-economic significance.	[An]	[5]

(10 × 1 = 10)